## 4.1 Introduction

Triangles are an effective tool for construction and architecture. They are used in the design of buildings and other structures for more strength and stablility. The knowledge of the properties of triangles is essential to understand its use in architecture. The triangle's use in architecture dates back more years than other common shapes like dome, arch, cylinder. Usage of triangles even predates the wheel. The most sturdy of the triangles are equilateral and isosceles; their symmetry aids in distributing weight.

This chapter is a continuation of properties of triangles that we studied in class VI.

**MATHEMATICS ALIVE - Geometry in Real Life**



*Electric Transformer*



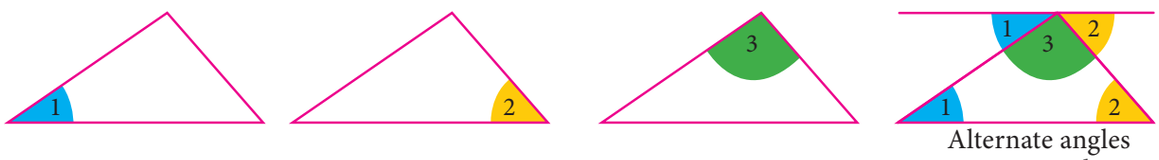
*Howrah Bridge*

## 4.2 Application of Angle Sum Property of Triangle

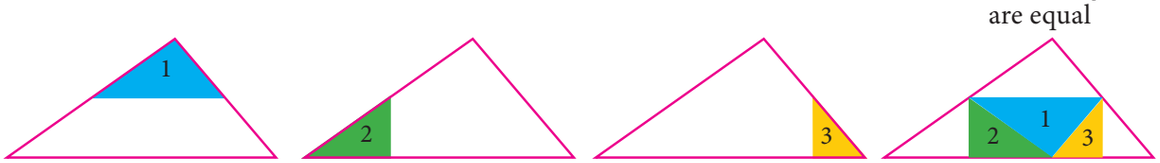
We are familiar with the angle sum property of a triangle which can be stated as the sum of all angles in a triangle is 180°. We can verify this by doing the following activity.

**Activity**

Draw any triangle and colour the angles. Check the property as shown below:

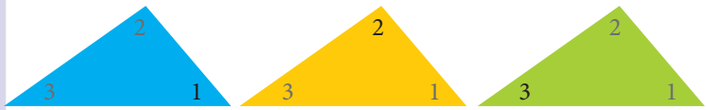



Alternate angles are equal




Flipping the three angles

Take three equiangular tirangles




Arranging the three triangles




From the above we have verified that the sum of three angles of any triangle is 180°



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